Pediatric anesthesia

Miller's anesthesia　Chapter 77

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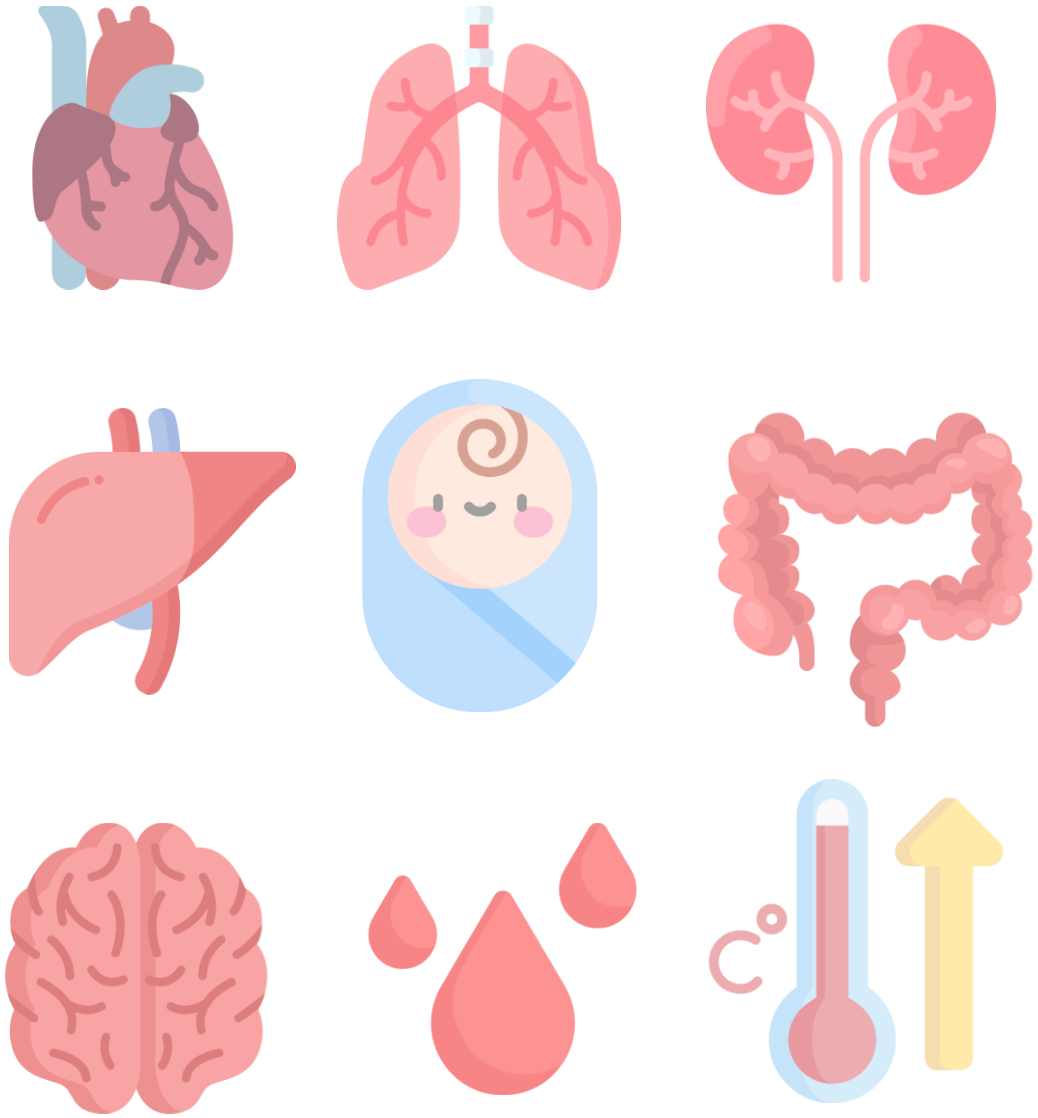

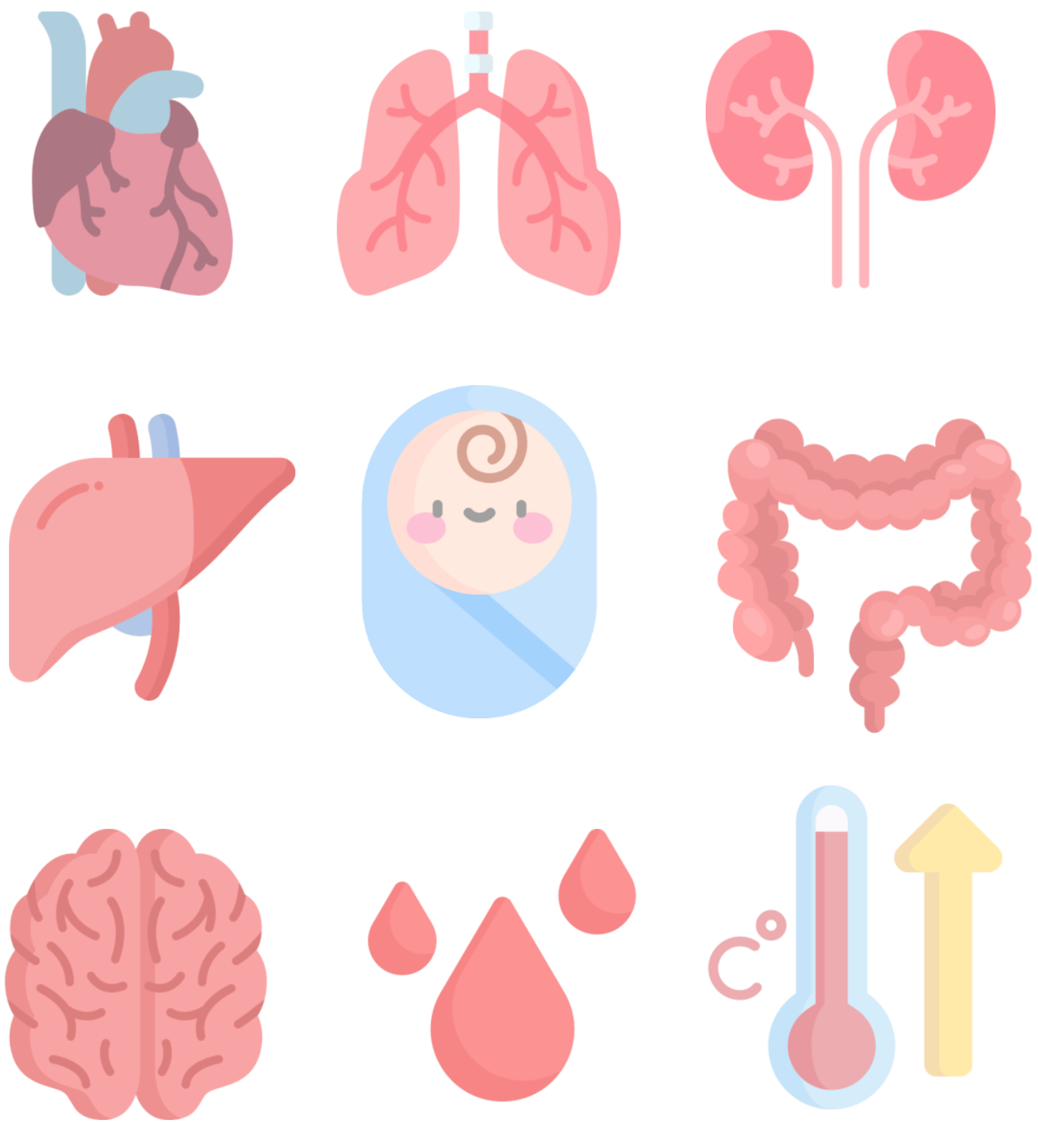

Organ System

Introduction

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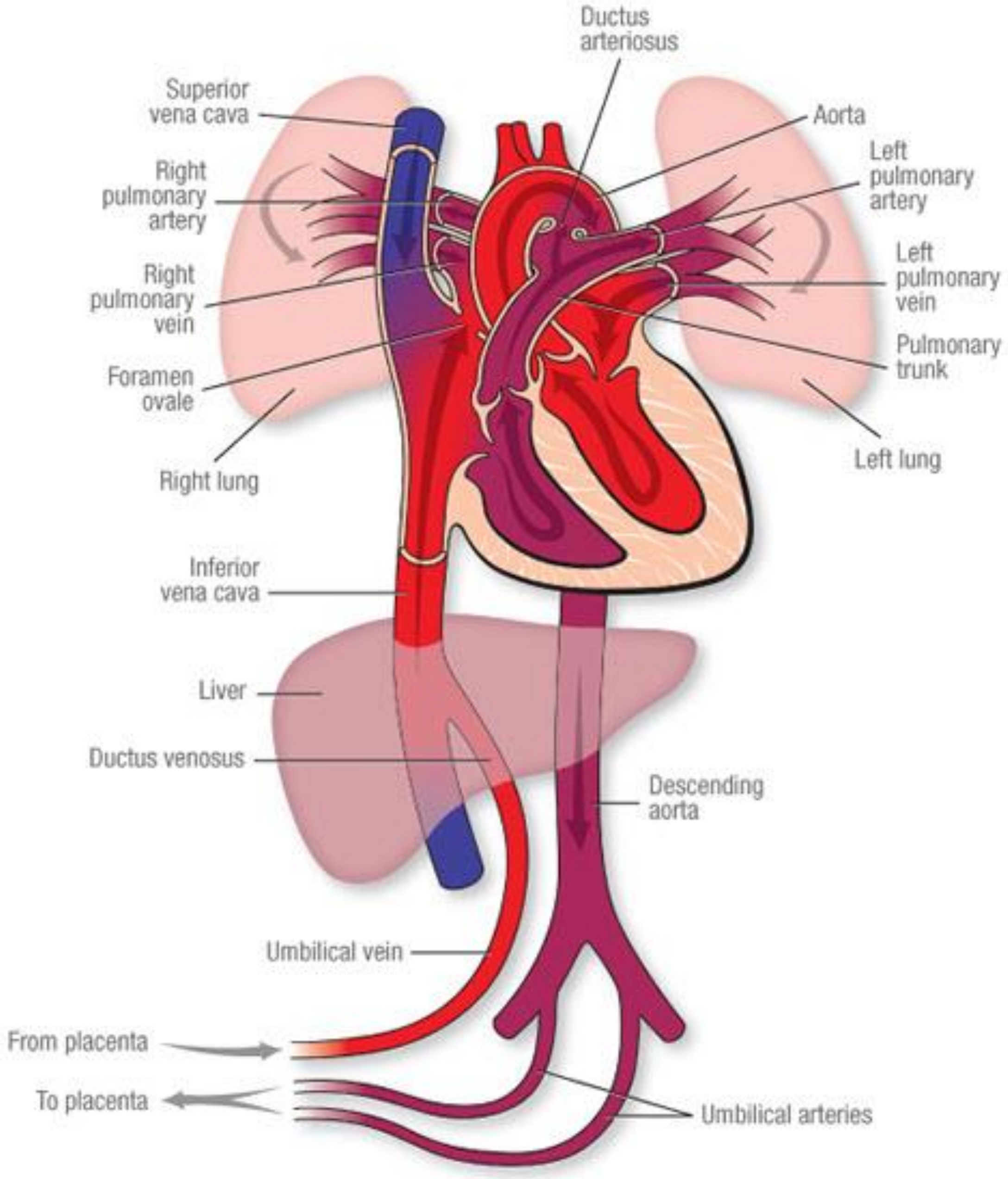
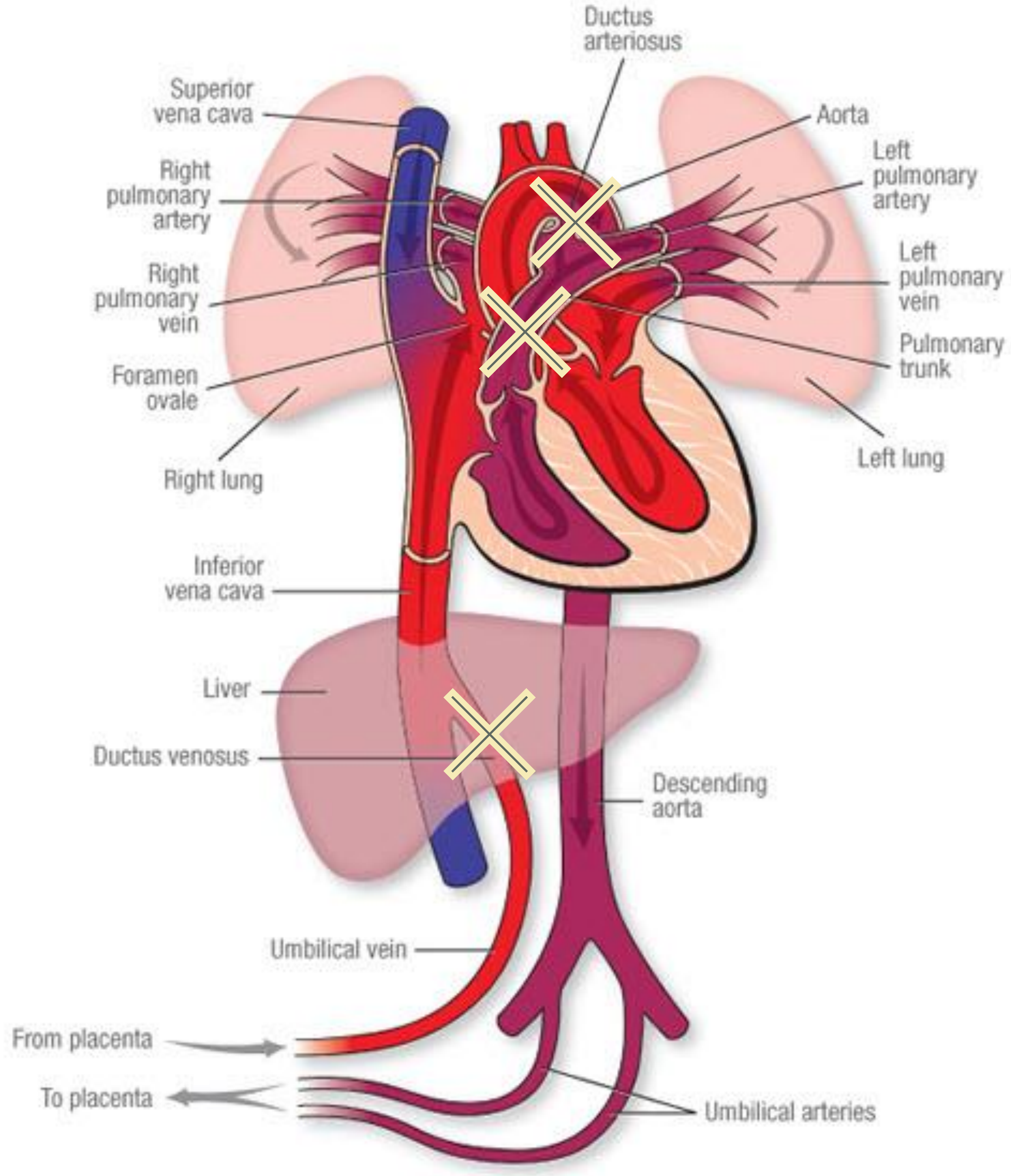
Circulation before birth

- 3 shunts before birth
  - Ductus venosus
  - Foramen ovale
  - Ductus arteriosus



Circulation after birth

- Placenta to lung, closure of 3 shunts
- SVR ⬆
- PVR ⬇ : day 1 to few years



Transitional circulation (flip-flop)

- Pathophysiology：

  PAP⬆ → opening of foramen ovale or ductus arteriosus → hypoxemia

- Etiology: **hypoxia, hypercapnia**, anesthesia induced SVR or PVR changes

- Risk factors：preterm, congenital heart disease, pulmonary disease(aspiration/pneumonia), infection, acidosis, hypothermia

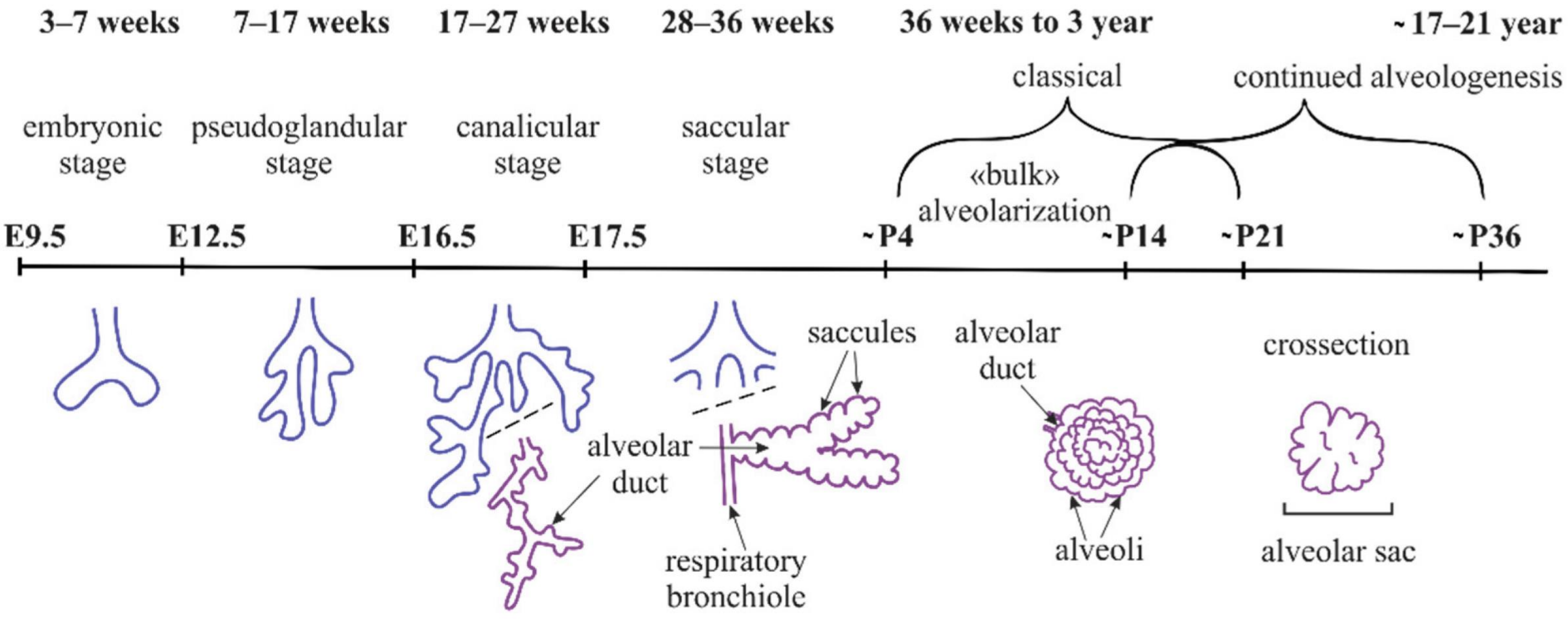
- Tx：**hyperventilation**, keep adequate body temperature

Cardiac output

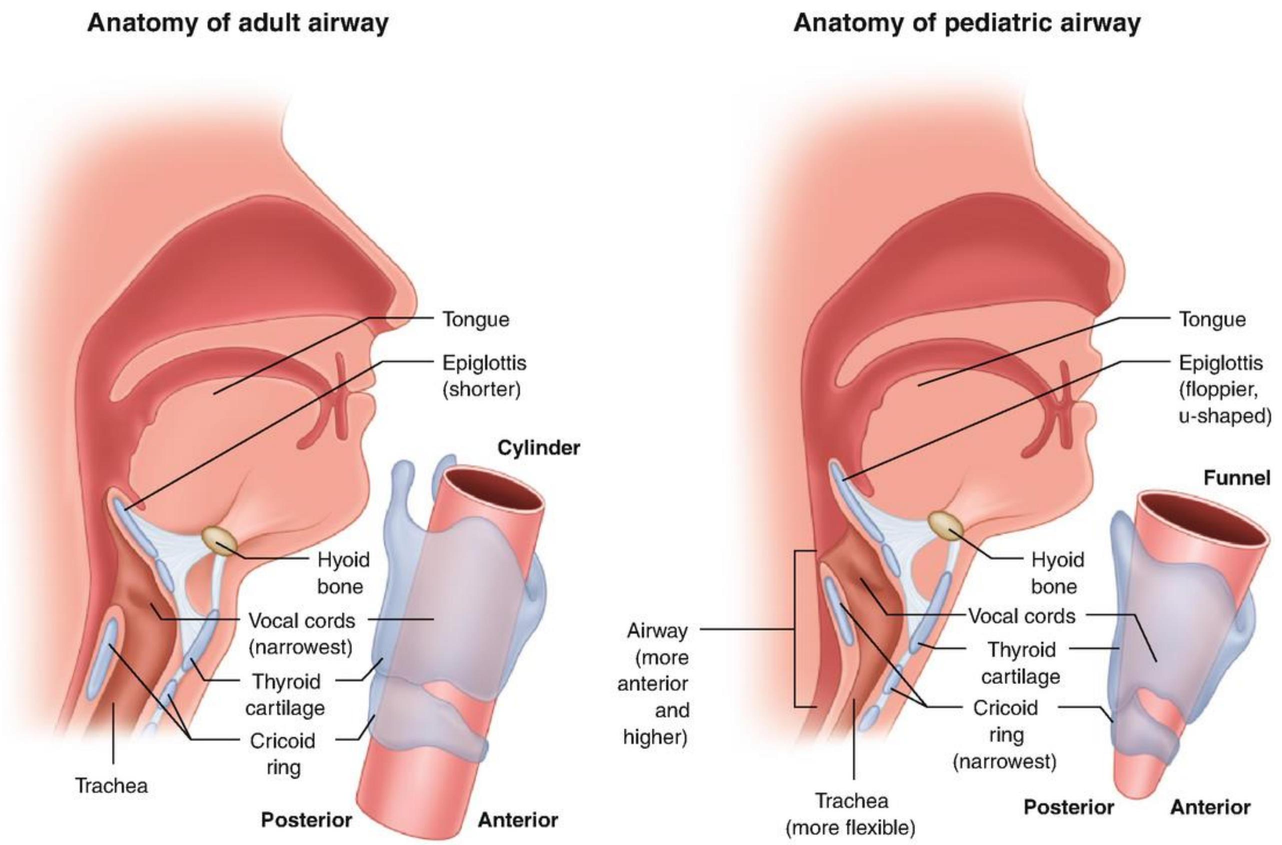
- Cardiac output = **Heart rate** x stroke volume
  - Hypoxia/ calcium channel blockade → **bradycardia** → asystole

- Characters of pediatric cardiac muscle
  - Less myofibril: water>collagen, connective tissue>contractile tissue
  - Few and immature mitochondria
  - Immature sarcoplasmic reticulum and T-tubule: depend on extracellular Ca
  - Dominant parasympathetic innervation

Lung embryology



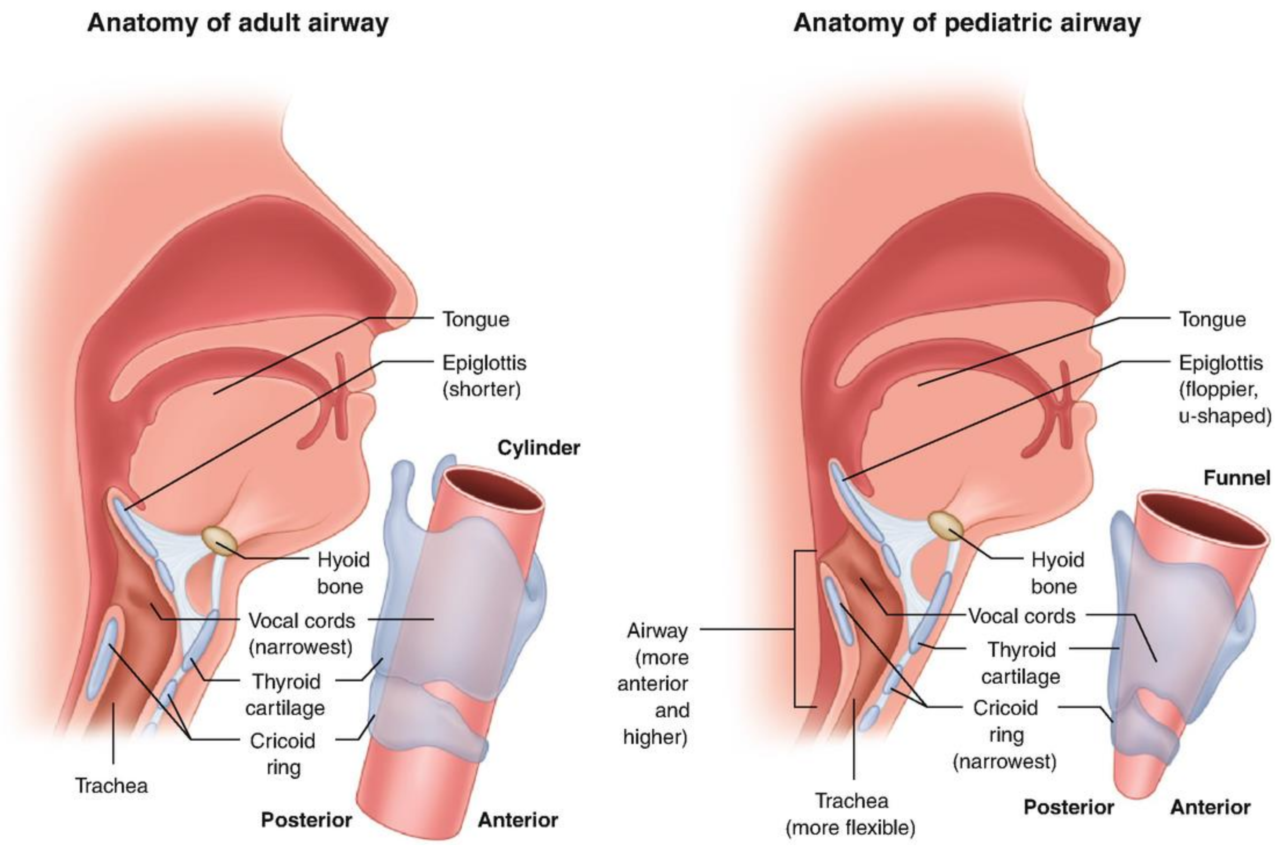
Airway structure 🫁



Upper airway

- Relatively large size of tongue
- Higher larynx
- Omega-shaped epiglottis

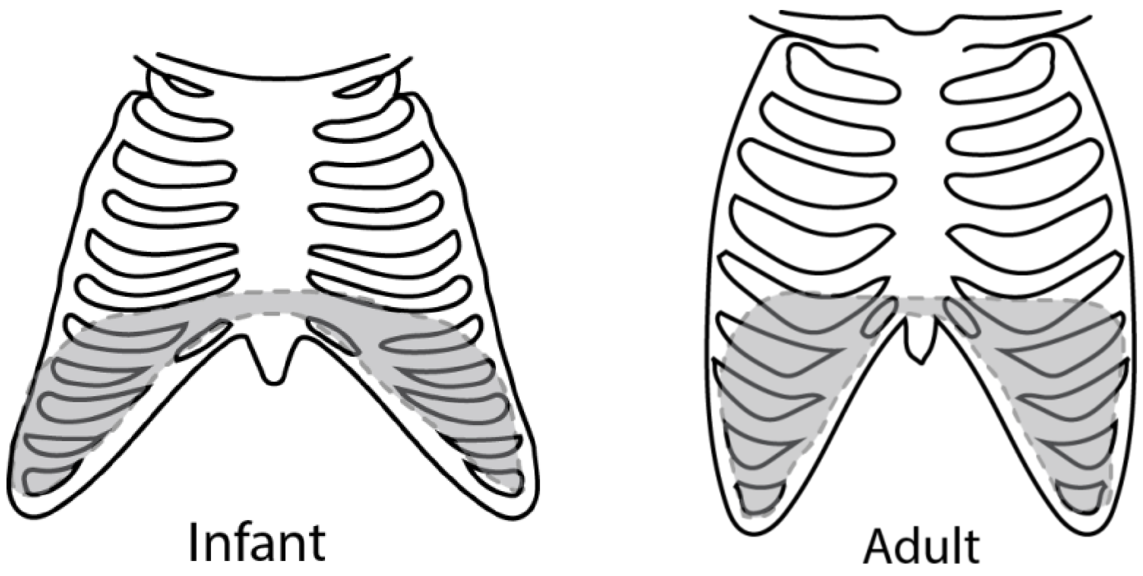
Airway structure 🫁



Lower airway

- Angled vocal cords
- Funnel shaped larynx

Airway structure 🫁



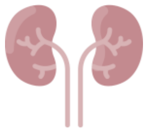
Chest wall

- Extremely compliant
- Horizontal ribs
- ⬇ Fatigue-resistant type I muscle fibers

Breathing

- Higher metabolism
  - ↑ oxygen consumption
  - rapid hypercapnia

- ↑ work of breath: ↑ **respiratory rate**
  - Easily airway obstruction
  - Cold stress

Kidney 

- Development
  - Preterm neonate: ↓ renal perfusion pressures, immature glomerulus and tubule
  - Term neonate: gradual maturation of glomerulus
  - **2 years: Complete maturation of renal function**

- Pharmacology
  - ↑ half-life of renal-secreted drugs

TABLE 77.3 Development of Glomerular Filtration Rate in the Term Newborn

Age	Glomerular Filtration Rate (mL/min/1.73 m^2 mean)	Range
1 day	24	3-38
2-8 days	38	17-60
10-22 days	50	32-68
37-95 days	58	30-86
1-2 years	115	95-135

Liver 

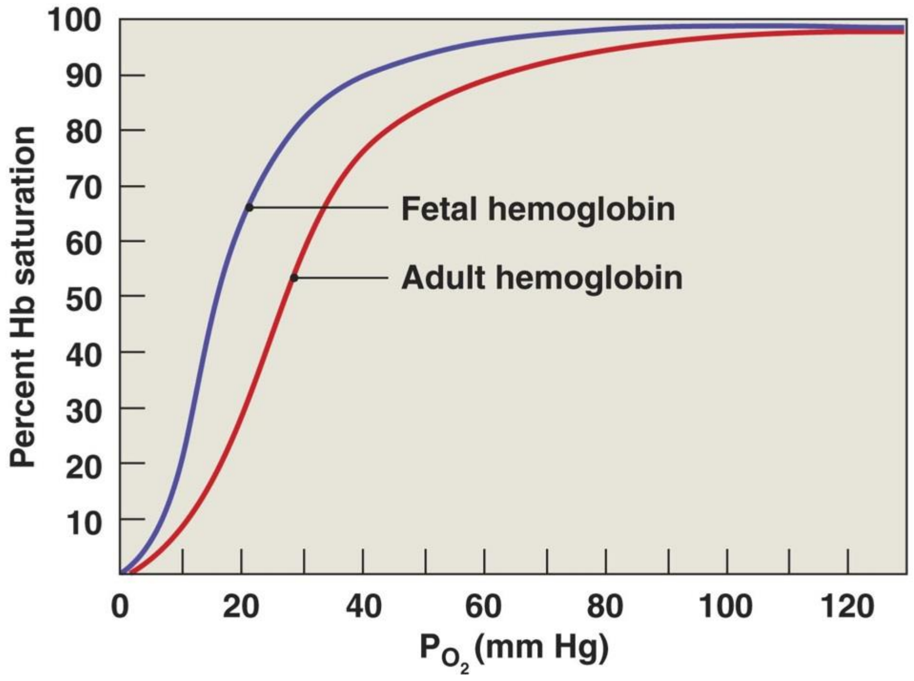
- Development
  - Term neonate: developed enzyme systems, but cannot metabolize materials
    - Phase 1 (oxidations-reductions and hydrolysis): variance among cytochrome 450 enzyme
    - Phase 2 (conjugation): maturation after 1y
  - **~ 2 years old: ↑ liver blood flow with mature enzyme system**
- Nutrition
  - Minimal glycogen store: easily hypoglycemia
  - Cannot afford large protein loads: ↓ albumin, ↑ unbound drugs, ↓ prothrombin

Gastrointestinal system

- Gastric juice
  - Day 1: alkalotic
  - **Day 2: normal pH range**
  - 4-5 months: coordinate swallowing with respiration

- GI abnormalities
  - Upper GI: **vomiting** and regurgitation
  - Lower GI: abdominal distention and **failure to pass meconium**

Hematologic system



- Hemoglobin for hypoxemic uterus
  - ⬆ Renal erythropoietin
  - ⬆ O2 affinity of fetal hemoglobin

→ high Hb level at birth, **physiologic anemia in 3 months**

→ fetal hemoglobin replaced by adult hemoglobin in 6 months

- Coagulation system
  - Normal amount and function after 6 months (platelet, fibrinogen and anticoagulants)

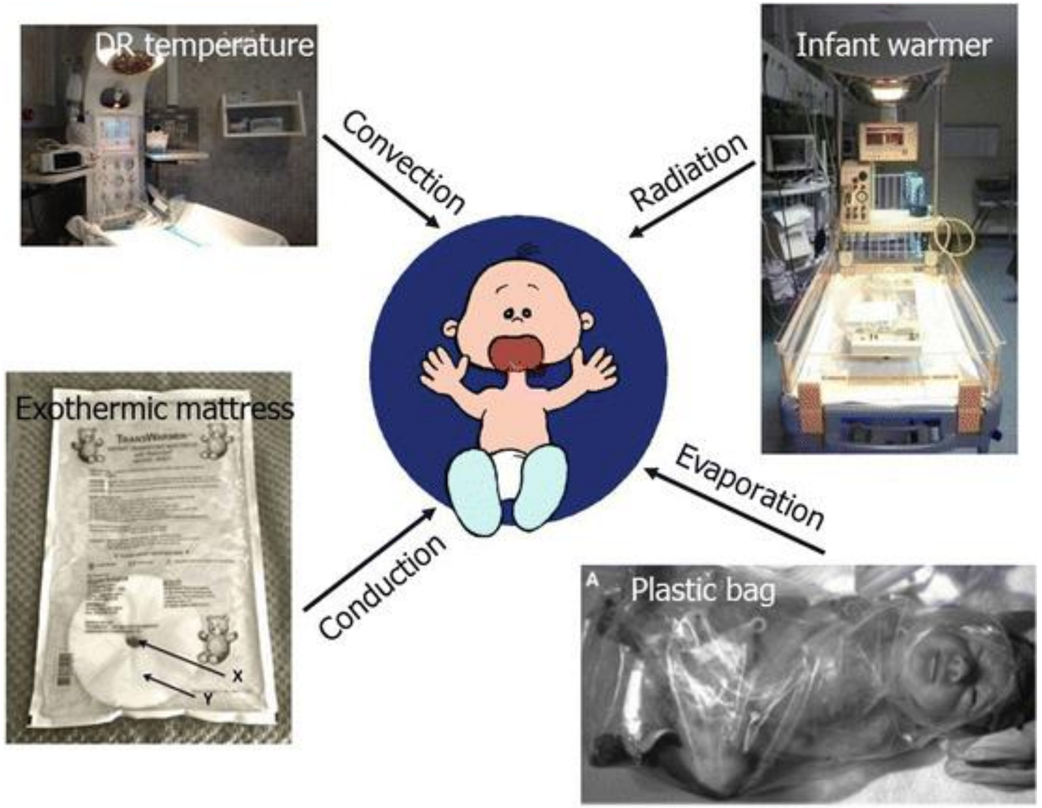
Central nervous system

- Development
  - GA 3-4w: neural tube
  - **3rd trimester ~ few years: brain growth spurt**

- Strong pain behavior derived from **protective withdrawl reflex**
  - Especially in preterm neonates
  - Affect brain activation, development of endogenous-gating system, somatosensory perception and pain sensitivity

Thermoregulation

- Vulnerable to hypothermia
  - ↑ ratio of body surface area to weight
  - ↓ skin thickness
  - ↓ fat store, ↓ shiver for cold stress



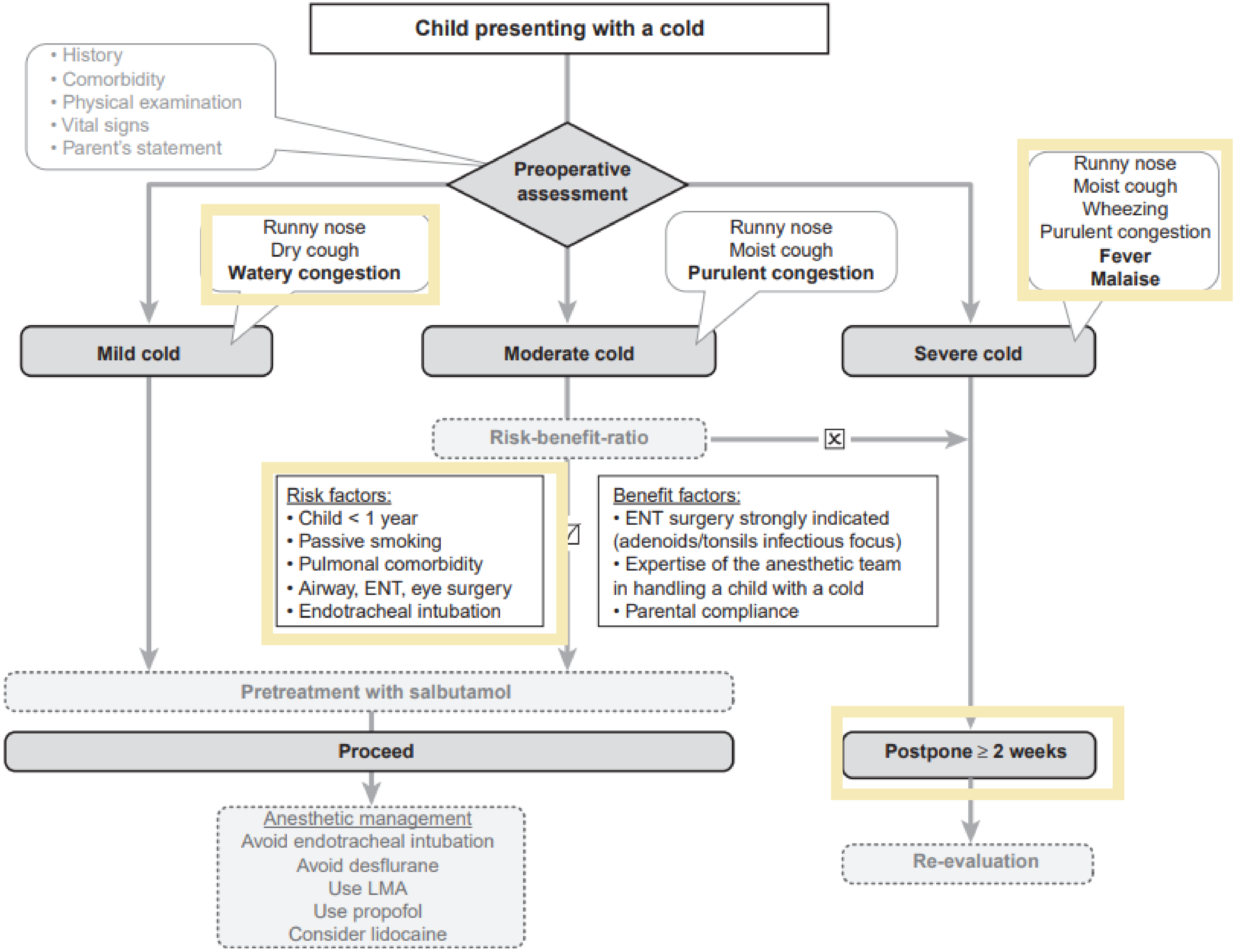
Anesthesia Management

Preoperative evaluation

- Underlying diseases: medication, previous history
- Physical examinations: airway assessment
- Lab and tests: EKG, chest x-ray
    - Harm of radiation?
    - <6m routine EKG, to rule out long QT syndrome
- Necessity of the operation

Child with URI

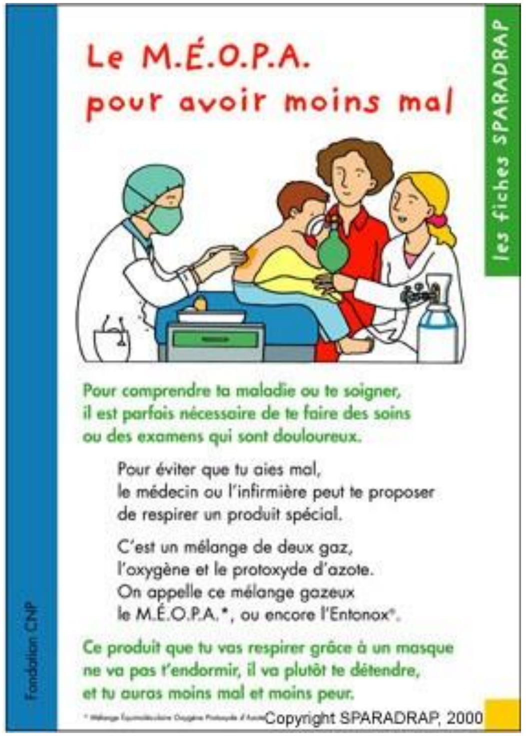
- Results: airway edema, secretion, **bronchial hypersensitivity**
- Risk assessment

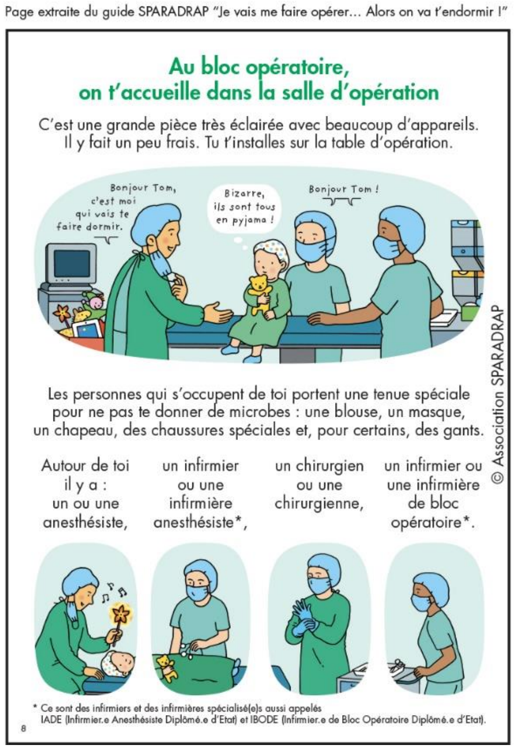


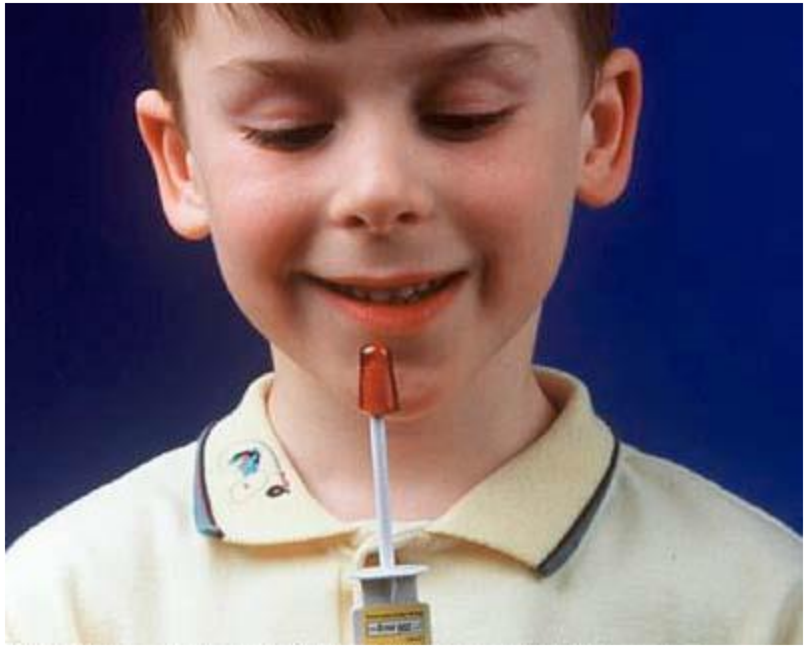
Child presenting with a cold
• History
• Comorbidity
• Physical examination
• Vital signs
• Parent's statement
Preoperative assessment
Runny nose
Dry cough
Watery congestion
Runny nose
Moist cough
Purulent congestion
Runny nose
Moist cough
Wheezing
Purulent congestion
Fever
Malaise
Mild cold
Moderate cold
Severe cold
Risk-benefit-ratio
Risk factors:
• Child < 1 year
• Passive smoking
• Pulmonal comorbidity
• Airway, ENT, eye surgery
• Endotracheal intubation
Benefit factors:
• ENT surgery strongly indicated (adenoids/tonsils infectious focus)
• Expertise of the anesthetic team in handling a child with a cold
• Parental compliance
Pretreatment with salbutamol
Proceed
Postpone ≥ 2 weeks
Anesthetic management
Avoid endotracheal intubation
Avoid desflurane
Use LMA
Use propofol
Consider lidocaine
Re-evaluation

Preoperative anxiolysis

- Differential diagnosis: **separation anxiety**, concerns about body damage…







Fentanyl lollipop

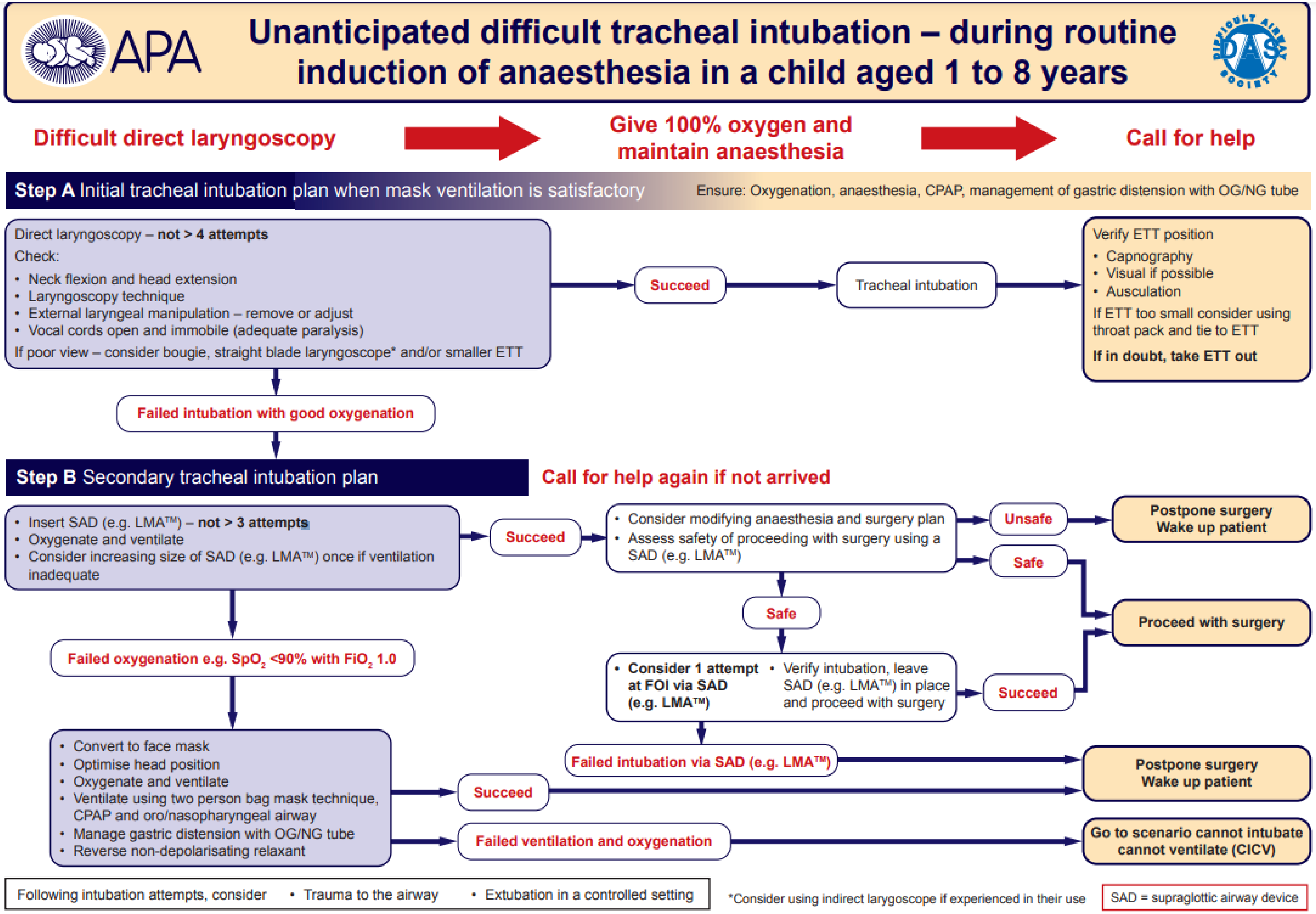
Fasting before surgery

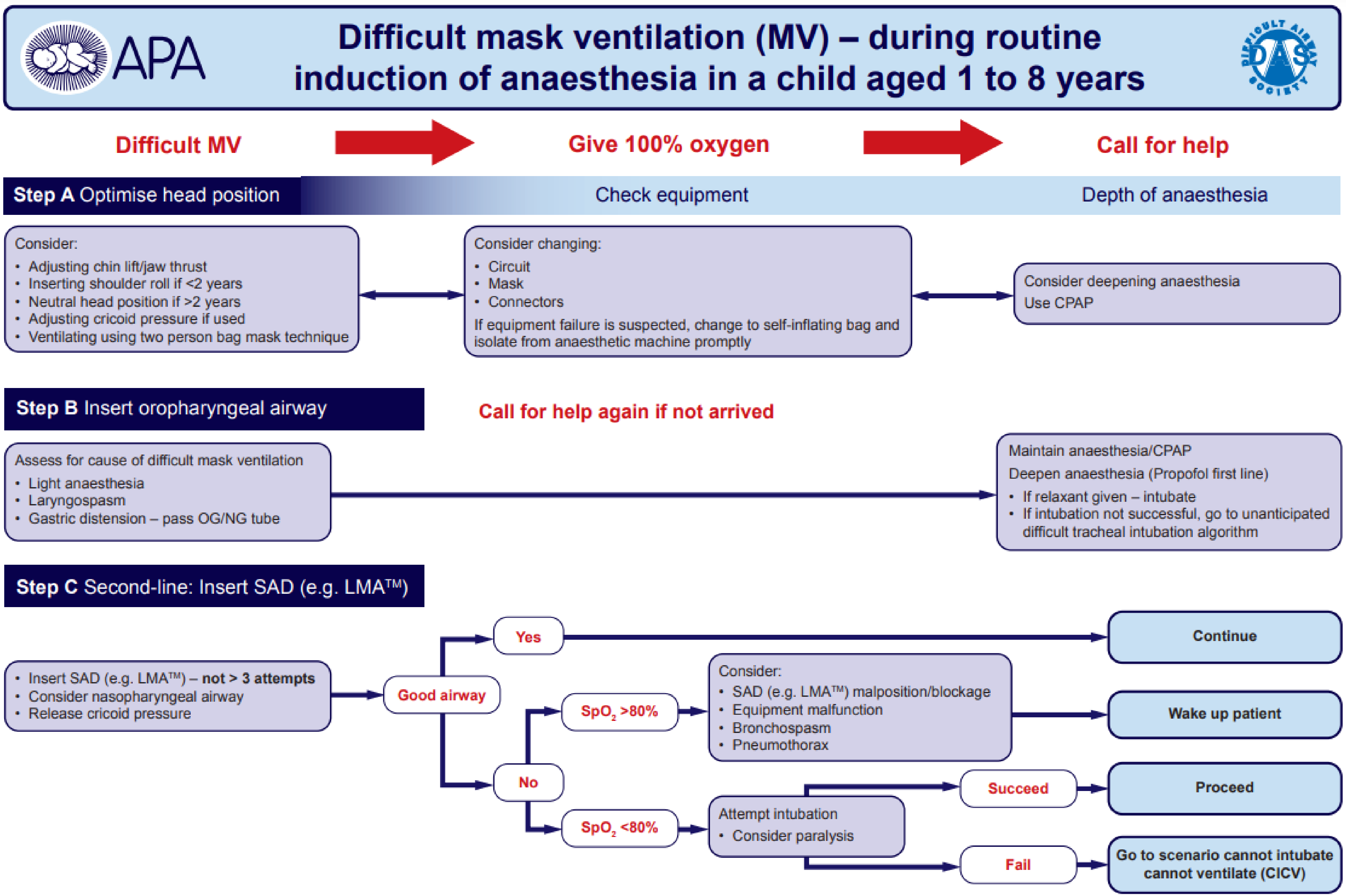
- Influences of longer fasting
  - Hypoglycemia
  - Metabolic acidosis
  - Dehydration
  - Cardiovascular instability
- **Low aspiration risk** of healthy child

→ New consensus: **1-hour-long fasting after clear fluid intake**

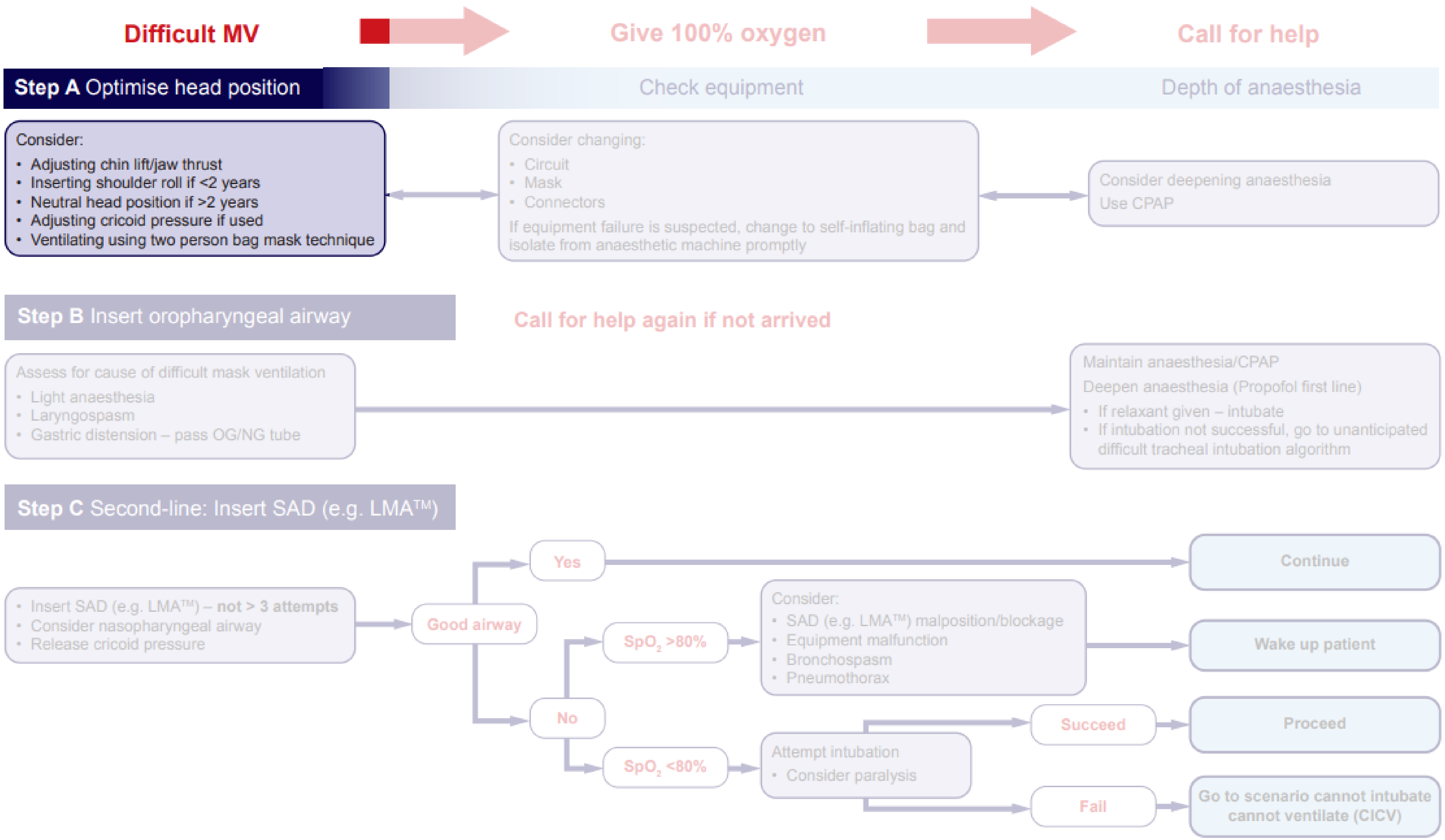
Airway management

- Mask ventilation
- Supraglottic airway device (LMA)
- Endotracheal tube
    - Uncuffed: ↓ cricoid cartilage damage, ↓ airway resistance, ↑ laryngospasm
    - **Cuffed**: ↓ airway trauma, accurate ETCO2


APA
Unanticipated difficult tracheal intubation – during routine induction of anaesthesia in a child aged 1 to 8 years
DAS
Difficult direct laryngoscopy
Give 100% oxygen and maintain anaesthesia
Call for help
Step A Initial tracheal intubation plan when mask ventilation is satisfactory
Ensure: Oxygenation, anaesthesia, CPAP, management of gastric distension with OG/NG tube
Direct laryngoscopy – not > 4 attempts
Check:
• Neck flexion and head extension
• Laryngoscopy technique
• External laryngeal manipulation – remove or adjust
• Vocal cords open and immobile (adequate paralysis)
If poor view – consider bougie, straight blade laryngoscope* and/or smaller ETT
Succeed
Tracheal intubation
Verify ETT position
• Capnography
• Visual if possible
• Auscultation
If ETT too small consider using throat pack and tie to ETT
If in doubt, take ETT out
Failed intubation with good oxygenation
Step B Secondary tracheal intubation plan
Call for help again if not arrived
• Insert SAD (e.g. LMA™) – not > 3 attempts
• Oxygenate and ventilate
• Consider increasing size of SAD (e.g. LMA™) once if ventilation inadequate
Succeed
• Consider modifying anaesthesia and surgery plan
• Assess safety of proceeding with surgery using a SAD (e.g. LMA™)
Unsafe
Postpone surgery
Wake up patient
Safe
Proceed with surgery
Safe
Failed oxygenation e.g. SpO2 <90% with FiO2 1.0
• Consider 1 attempt at FOI via SAD (e.g. LMA™)
• Verify intubation, leave SAD (e.g. LMA™) in place and proceed with surgery
Succeed
Failed intubation via SAD (e.g. LMA™)
• Convert to face mask
• Optimise head position
• Oxygenate and ventilate
• Ventilate using two person bag mask technique, CPAP and oro/nasopharyngeal airway
• Manage gastric distension with OG/NG tube
• Reverse non-depolarising relaxant
Succeed
Postpone surgery
Wake up patient
Failed ventilation and oxygenation
Go to scenario cannot intubate cannot ventilate (CICV)
Following intubation attempts, consider • Trauma to the airway • Extubation in a controlled setting
*Consider using indirect larygoscope if experienced in their use
SAD = supraglottic airway device


APA
Difficult mask ventilation (MV) – during routine induction of anaesthesia in a child aged 1 to 8 years
DAS DIFFICULT AIRWAY SOCIETY
Difficult MV
Give 100% oxygen
Call for help
Step A Optimise head position
Check equipment
Depth of anaesthesia
Consider:
• Adjusting chin lift/jaw thrust
• Inserting shoulder roll if <2 years
• Neutral head position if >2 years
• Adjusting cricoid pressure if used
• Ventilating using two person bag mask technique
Consider changing:
• Circuit
• Mask
• Connectors
If equipment failure is suspected, change to self-inflating bag and isolate from anaesthetic machine promptly
Consider deepening anaesthesia
Use CPAP
Step B Insert oropharyngeal airway
Call for help again if not arrived
Assess for cause of difficult mask ventilation
• Light anaesthesia
• Laryngospasm
• Gastric distension – pass OG/NG tube
Maintain anaesthesia/CPAP
Deepen anaesthesia (Propofol first line)
• If relaxant given – intubate
• If intubation not successful, go to unanticipated difficult tracheal intubation algorithm
Step C Second-line: Insert SAD (e.g. LMA™)
• Insert SAD (e.g. LMA™) – not > 3 attempts
• Consider nasopharyngeal airway
• Release cricoid pressure
Good airway
Yes
No
Continue
SpO2 >80%
Consider:
• SAD (e.g. LMA™) malposition/blockage
• Equipment malfunction
• Bronchospasm
• Pneumothorax
Wake up patient
SpO2 <80%
Attempt intubation
• Consider paralysis
Succeed
Proceed
Fail
Go to scenario cannot intubate cannot ventilate (CICV)

Difficult mask ventilation (MV) – during routine induction of anaesthesia in a child aged 1 to 8 years



Difficult mask ventilation (MV) – during routine induction of anaesthesia in a child aged 1 to 8 years

APA | DAS

Difficult MV → Give 100% oxygen → Call for help

Step A Optimise head position | Check equipment | Depth of anaesthesia

Consider:
- Adjusting chin lift/jaw thrust
- Inserting shoulder roll if <2 years
- Neutral head position if >2 years
- Adjusting cricoid pressure if used
- Ventilating using two person bag mask technique

Consider changing:
- Circuit
- Mask
- Connectors

If equipment failure is suspected, change to self-inflating bag and isolate from anaesthetic machine promptly

Consider deepening anaesthesia
Use CPAP

Step B Insert oropharyngeal airway

Call for help again if not arrived

Assess for cause of difficult mask ventilation
- Light anaesthesia
- Laryngospasm
- Gastric distension – pass OG/NG tube

Maintain anaesthesia/CPAP
Deepen anaesthesia (Propofol first line)
- If relaxant given – intubate
- If intubation not successful, go to unanticipated difficult tracheal intubation algorithm

Step C Second-line: Insert SAD (e.g. LMA™)

- Insert SAD (e.g. LMA™) – **not > 3 attempts**
- Consider nasopharyngeal airway
- Release cricoid pressure

Good airway

Yes → Continue

No → SpO_2 >80% → Consider:
- SAD (e.g. LMA™) malposition/blockage
- Equipment malfunction
- Bronchospasm
- Pneumothorax

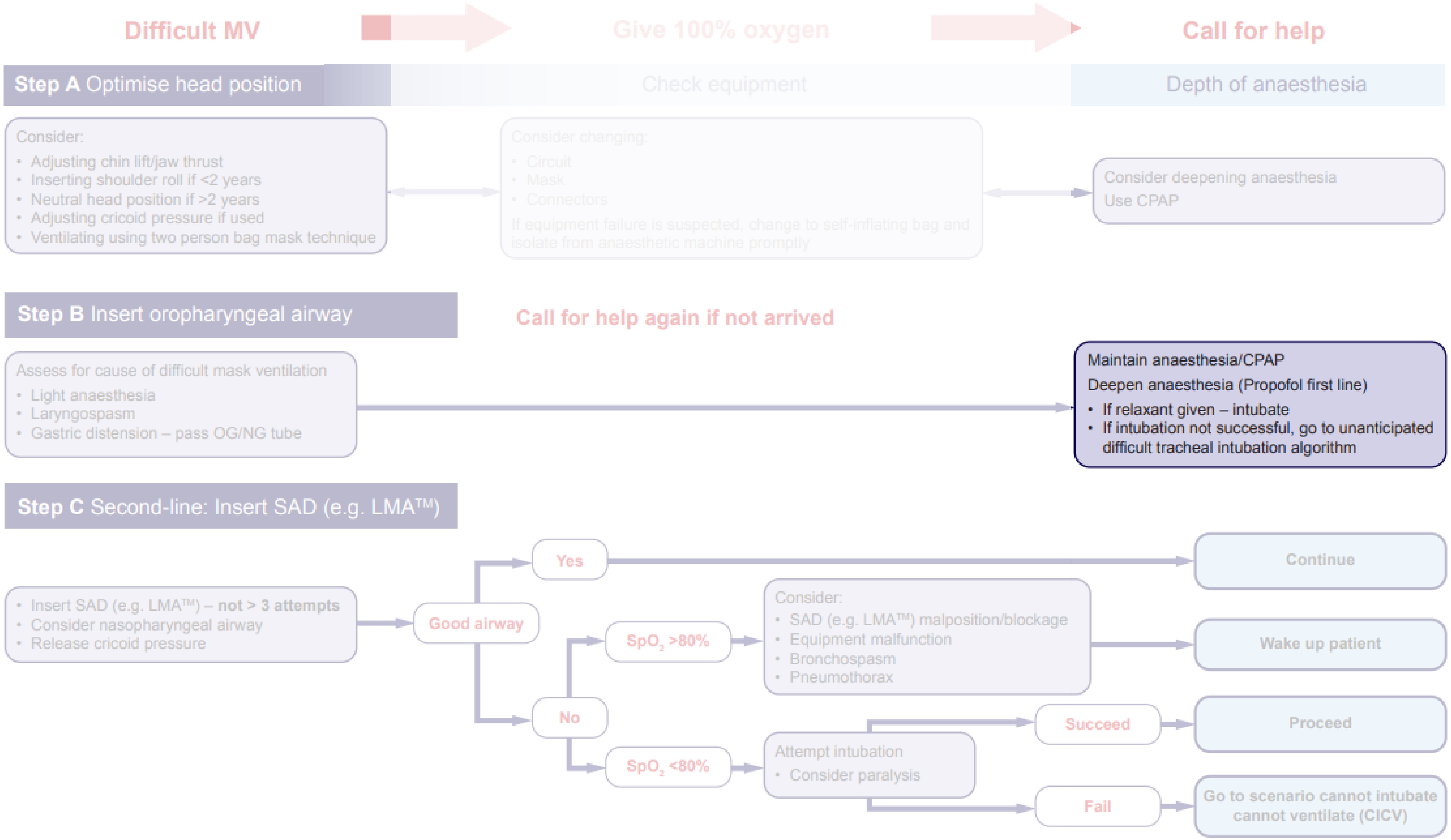
→ Wake up patient

No → SpO_2 <80% → Attempt intubation
- Consider paralysis

Succeed → Proceed

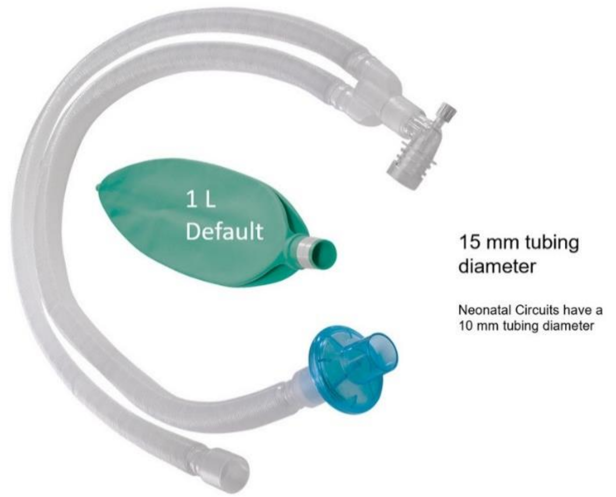
Fail → Go to scenario cannot intubate cannot ventilate (CICV)

Difficult mask ventilation (MV) – during routine induction of anaesthesia in a child aged 1 to 8 years



Ventilation strategies

- **Ventilator induced lung injury (VILI)**
  - Age dependent susceptibility
  - Ideal tidal volume: **6-10ml/kg**
- PEEP: around **5cmH2O**
- Dead space: ⬆dead space/tidal volume ratio → ⬆PaCO2



Fluid management

Maintenance requirement	**4-2-1 rule**
+	
Preoperative deficits	4-2-1 rule (1st hr: half, 2nd hr: half) **15-25ml/kg during 1st hour**
+	
Ongoing loss	1-15ml/kg/hr

Fluid management

- **Isotonic crystalloid** > hypotonic crystalloid
    - Avoid cerebral edema
    - Regular monitor electrolytes

- Dextrose water ?
    - Pre-operative hypoglycemia risk 0-2.5%
    - If hyperglycemia →lactic acidosis, electrolyte abnormalities, poor neurological outcome
    - **1-2.5% dextrose water** in high risk groups

Conclusion

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